Noise-Adaptive Tolls in Atomic Congestion Games

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Abstract: We consider a congestion game with a finite number of agents selecting paths according to a logit dynamics with fixed noise level. We propose an iterative distributed tolling scheme and show that, for every fixed noise level, the expected total travel time can be made arbitrarily close to the system optimum by taking sufficiently many toll updates and a sufficiently large population.

Keywords: Transportation Networks, Control of Networks, Multi-agent systems, Game Theories, Cyber-physical urban systems

1. INTRODUCTION

We consider a congestion game with a finite number of agents selecting paths according to a logit dynamics with fixed noise level. When the population is infinitely large, marginal cost tolls can restore the social optimum under the assumption of perfect rationality (Beckmann et al. (1956)), but these assumptions rarely hold. In fact, in presence of noise, as the number of agents grows large, the invariant distribution of the logit dynamics concentrates on perturbed equilibria, which are in general suboptimal (Sandholm (2011)). In this work, we establish an iterative distributed algorithm for toll design in congestion games to minimize the expected total travel time on the network for every noise level, when the number of players is large but finite. This work complements the results in Park and Leonard (2025), where an optimal incentive mechanism was provided for every noise level in contractive games with a continuum of users.

Notation: We let Δ, $\Delta_n = \Delta \cap \frac{1}{n}\mathbb{Z}_+^m$, and $\mathrm{int}(\Delta)$, denote the m-dimensional simplex, its discrete counterpart — whose size may be deduced from the context— and the interior of Δ, respectively. The Dirac delta centered in θ is denoted $\delta^\theta(\cdot)$, while u^i denotes the vector with 1 in the i-th entry and 0 elsewhere.

2. PROBLEM SETTING

We model the network by a directed multigraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. We consider a single origin and destination nodes o, d in $\mathcal{V}$, with d reachable from o, and let Γ denote the non-empty set of $o-d$ paths. We define the link-path incidence matrix A in $\{0,1\}^{\mathcal{E}\times\Gamma}$ with entries $A_{e\gamma} = 1$ if and only if e belongs to γ. We consider n identical players, each choosing a path in Γ, and collect the empirical path frequencies into the state vector x in Δ_n, with x_γ denoting the fraction of players choosing path γ. The cost of γ is

$$c_\gamma^\omega(x) = \sum_{e\in\mathcal{E}} A_{e\gamma}(\tau_e((Ax)_e) + \omega_e((Ax)_e)), \qquad (1)$$

where, for every link e in $\mathcal{E}$, $\tau_e : \mathbb{R}_+ \to \mathbb{R}$ is a continuously differentiable, strictly increasing, positive, and convex delay function that accounts for congestion, and $\omega_e : \mathbb{R}_+ \to \mathbb{R}_+$ is a monetary *toll*. This setting defines a population game.

The population state evolves by the following discrete-time Markov chain $X(t)$ on Δ_n, called *logit dynamics*. At each time step, an agent is selected uniformly at random and chooses a random path according to a logit probability distribution. That is, for every two states x, y in Δ_n and two paths i, j in Γ, the transition probability is

$$P_{xy}^{(n)} = \begin{cases} x_i \dfrac{e^{-\eta^{-1}c_j^\omega(y)}}{\sum_{k\in\Gamma} e^{-\eta^{-1}c_k^\omega(x+\frac{1}{n}(u^k-u^i))}} & \text{if } y = x + \dfrac{1}{n}(u^j - u^i) \\ 0 & \text{otherwise}, \end{cases}$$

where $\eta > 0$ is the noise level. The Markov chain $X(t)$ is irreducible for every noise level. Hence, it admits a unique invariant distribution π_n^ω that depends on the tolls ω.

The goal of the planner is to design a vector of tolls to minimize the expected total travel time, i.e.,

$$\inf_{\{\omega_e:\mathbb{R}_+\to\mathbb{R}_+\}_{e\in\mathcal{E}}} \mathbb{E}_{\pi_n^\omega}[\phi(X)], \qquad (2)$$

where $\phi(x) : \Delta \to \mathbb{R}_+$ is defined by

$$\phi(x) = \sum_{\gamma\in\Gamma} x_\gamma \sum_{e\in\mathcal{E}} A_{e\gamma}\tau_e((Ax)_e). \qquad (3)$$

To conclude this section, we introduce the system-optimum flows, i.e., the minimizers of the total travel time

$$\mathcal{X}^* = \arg\min_{x\in\Delta} \phi(x).$$

3. OPTIMAL TOLLS

In this section, we shall provide a toll vector under which the expected total travel time approaches the system optimum as the number of players grows large, working under the following assumption.

Assumption 1. The transportation network $\mathcal{G}$ has the property that each path contains at least one exclusive link, i.e., there exists an injective map $l : \Gamma \to \mathcal{E}$ such that

$$A_{l(\gamma)\gamma} = 1\,, \qquad A_{l(\gamma)\gamma'} = 0\,, \qquad \forall \gamma' \neq \gamma.$$

Remark 1. Under Assumption 1, the link-path incidence matrix A has full column rank and is therefore injective. Consequently, ϕ is strictly convex on Δ and admits a unique minimizer, i.e., $\mathcal{X}^* = \{x^*\}$ is a singleton.

In the limit of infinitely large population, the feedback marginal cost tolls

$$\tilde{\omega}_e(x) = (Ax)_e \tau_e'((Ax)_e)\,, \quad e \in \mathcal{E}\,,$$

are known to achieve optimality. However, in presence of noise, these tolls are no longer optimal (Sandholm (2011)). In this work, we combine marginal cost tolls with noise-dependent tolls that address the distortion introduced by the noise. Such tolls $\bar{\omega}(\theta)$ are parametrized by θ in $\mathrm{int}(\Delta)$ and are defined by

$$\bar{\omega}_e(\theta) = \begin{cases} -\eta \log(\theta_{l^{-1}(e)})\,, & \text{if } e = l(\gamma) \text{ for some } \gamma \in \Gamma \\ 0 & \text{otherwise}\,. \end{cases}$$

The idea is to set a toll $-\eta \log(\theta_\gamma)$ on every path γ, and to decompose such tolls over the links so that, for every path γ, the toll is charged over the exclusive link $l(\gamma)$.

We now propose an iterative distributed algorithm for updating θ in such a way that the toll vector

$$\omega_e^\theta(x) = \tilde{\omega}_e(x) + \bar{\omega}_e(\theta)\,, \quad \forall e \in \mathcal{E}\,, \tag{4}$$

yields an expected total travel time arbitrarily close to the system optimum when the number of iterations and the population size are sufficiently large.

The parameter θ is updated through a discrete-time dynamical system that evolves on a much slower time scale than the underlying Markov chain. Let π_n^θ denote the invariant distribution of the Markov chain $X(t)$ with tolls (4). Then, let $\psi : \Delta \to \Delta$ defined by

$$\psi_n(\theta) = \mathbb{E}_{\pi_n^\theta}[X] = \sum_{x \in \Delta_n} x \pi_n^\theta(x)\,,$$

and consider the discrete-time dynamical system

$$\theta^{k+1} = \psi_n(\theta^k), \quad k \geq 0\,, \tag{5}$$

i.e., θ^{k+1} is the average state observed with tolls ω^{θ^k}.

Theorem 1. Consider the tolls ω^θ defined in (4). For every initial condition θ^0 in $\mathrm{int}(\Delta)$ and for every $\epsilon > 0$, there exists k_ϵ in $\mathbb{N}$ and, for every $k > k_\epsilon$, an integer $n^*_{\epsilon,k,\theta^0}$ in $\mathbb{N}$ such that

$$\mathbb{E}_{\pi_n^{\theta^k}}\Big[\phi(X)\Big] - \inf_{\pi \in \Delta_n} \mathbb{E}_\pi\Big[\phi(X)\Big] < \epsilon.$$

for every $n > n^*_{\epsilon,k,\theta^0}$.

Proof. The proof is deferred to Appendix C.

4. EXAMPLE

In order to corroborate our theoretical results, we provide a numerical example. Consider the Wheatstone network in

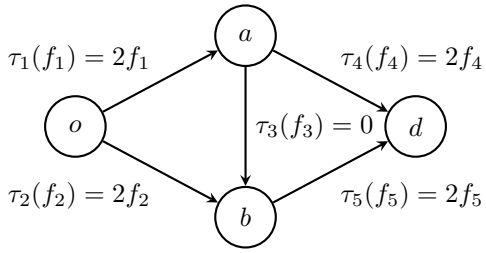


Fig. 1. Wheatstone network.

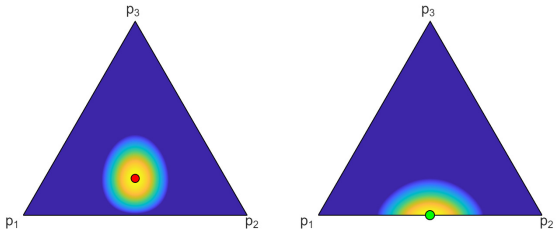


Fig. 2. Invariant distribution of the Markov Chain with $n = 500$ players and $\eta = 1$. Left: Marginal tolls only, in red the perturbed equilibrium. Right: Marginal + noise dependent tolls, in green the social optimum.

Figure 1, with path set $\Gamma = \{(o, a, d), (o, b, d), (o, a, b, d)\}$, which we shall denote by p_1, p_2, and p_3, respectively.

We consider a congestion game with $n = 500$ players, and compute the invariant distribution with noise level $\eta = 1$. As shown in the left plot of Figure 2, with the employment of only marginal tolls the invariant distribution concentrates around a perturbed equilibrium. The right plot of Figure 2 reports the invariant distribution, under tolls as in (4), which concentrates around the system-optimum $(1/2, 1/2, 0)$.

5. CONCLUSION

We proposed an iterative distributed tolling scheme that steers the invariant distribution of the noisy best response toward the system-optimum. Unlike classical marginal-cost pricing, the proposed mechanism remains effective for every noise level. Future work aim to extend these results to the case of arbitrary network topologies.

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Appendix A. PROPOSITION 1

In order to aid with the study of the Markov Chain, we introduce the function $\Psi : \Delta \to \Delta$ defined by

$$\Psi(\theta) = \arg\min_{x\in\Delta} \left\{ \eta^{-1}\phi(x) + D_{KL}(x||\theta) \right\} ,$$

which prompts the study of the dynamical system

$$\theta^{k+1} = \Psi(\theta^k) . \tag{A.1}$$

This map approximates (5) when the number of players grows infinitely large. Its analysis is instrumental to the proof of Theorem 1.

Proposition 1. Every $x^* \in \mathcal{X}^*$ is a fixed point of (A.1). Moreover, for every $\theta^0 \in \text{int}(\Delta)$ the sequence $\{\theta^k\}_{k\geq 0}$ admits a limit in $\mathcal{X}^*$.

Proof. First, we shall prove that ϕ is a Lyapunov function for (A.1). Since

$$\theta^{k+1} = \arg\min_{x\in\Delta} \Big\{ \eta^{-1}\phi(x) + D_{KL}(x\|\theta^k) \Big\} ,$$

and $D_{KL}(\theta^k||\theta^k) = 0$, it holds true that

$$\eta^{-1}\phi(\theta^{k+1}) + D_{KL}(\theta^{k+1}||\theta^k) \leq \eta^{-1}\phi(\theta^k) . \tag{A.2}$$

Moreover, since D_{KL} is non-negative, (A.2) implies

$$\phi(\theta^{k+1}) \leq \phi(\theta^k) .$$

Hence, ϕ is non-increasing along the trajectory and is a Lyapunov function for (A.1). Furthermore, since ϕ is non-negative by construction, the sequence $\{\phi(\theta^k)\}_{k\geq 0}$ admits a limit ℓ in $\mathbb{R}$, i.e.,

$$\lim_{k\to+\infty} \phi(\theta^k) = \ell .$$

We now show that every element of $\mathcal{X}^*$ is a fixed point of (A.1). Let x^* in $\mathcal{X}^*$. Since $\phi(x^*) \leq \phi(x)$ for every x in Δ, we have that

$$\eta^{-1}\phi(x^*) + D_{KL}(x^*||x^*) \leq \eta^{-1}\phi(x) + D_{KL}(x||x^*)$$

and the inequality is strict if and only if $x \neq x^*$. This shows that

$$\Psi(x^*) = x^* .$$

So far, we have proven that $\phi(x)$ is a Lyapunov function for (A.1) and that every x^* in $\mathcal{X}^*$ is a fixed point for (A.1). We now have to prove that the sequence θ^k admits a limit in $\mathcal{X}^*$. Towards this goal, we shall use another Lyapunov function for system (A.1). Recall that, given θ^k in $\text{int}(\Delta)$, we have

$$\theta^{k+1} = \arg\min_{x\in\Delta} \{ \eta^{-1}\phi(x) + D_{KL}(x||\theta^k) \} .$$

Hence, since the derivative of $D_{KL}(x\|\theta)$ diverges as x approaches the boundary of Δ, the minimum θ^{k+1} belongs to $\text{int}(\Delta)$. Hence, the first-order optimality conditions imply that, for every z in Δ,

$$\left\langle \eta^{-1}\nabla\phi(\theta^{k+1}) + \nabla D_{KL}(\theta^{k+1}\|\theta^k),\, z - \theta^{k+1} \right\rangle = 0 , \tag{A.3}$$

Moreover, since ϕ is convex, it holds that

$$\phi(\theta^{k+1}) - \phi(z) \leq \left\langle \nabla\phi(\theta^{k+1}),\, \theta^{k+1} - z \right\rangle . \tag{A.4}$$

Multiplying (A.4) by $\eta^{-1} > 0$ and using (A.3), we obtain

$$\eta^{-1}\big(\phi(\theta^{k+1}) - \phi(z)\big) \leq \left\langle \nabla D_{KL}(\theta^{k+1}\|\theta^k),\, z - \theta^{k+1} \right\rangle . \tag{A.5}$$

Now, note that, for every $\gamma \in \Gamma$, and $\theta, \theta' \in \Delta$,

$$\frac{\partial}{\partial\theta_\gamma} D_{KL}(\theta\|\theta') = \frac{\partial}{\partial\theta_\gamma} \sum_{a\in\Gamma} \theta_a \log\Big(\frac{\theta_a}{\theta'_a}\Big) = \log\Big(\frac{\theta_\gamma}{\theta'_\gamma}\Big) + 1 .$$

We can then expand the right-hand side of (A.5) and get

$$\begin{aligned}
\langle \nabla D_{KL}(\theta^{k+1}||\theta^k), z - \theta^{k+1} \rangle &= \sum_{\gamma\in\Gamma} (z_\gamma - \theta^{k+1}_\gamma) \log\Big(\frac{\theta^{k+1}_\gamma}{\theta^k_\gamma}\Big) + \\
&\quad + \sum_{\gamma\in\Gamma} (z_\gamma - \theta^{k+1}_\gamma) \\
&= \sum_{\gamma\in\Gamma} (z_\gamma - \theta^{k+1}_\gamma) \log\Big(\frac{\theta^{k+1}_\gamma}{\theta^k_\gamma}\Big) ,
\end{aligned}$$

where the second equality follows from the fact that z and θ belong to Δ. Notice that this can be equivalently rewritten as

$$\begin{aligned}
&\sum_{\gamma\in\Gamma} (z_\gamma - \theta^{k+1}_\gamma) \log\Big(\frac{\theta^{k+1}_\gamma}{\theta^k_\gamma}\Big) \\
&= \sum_{\gamma\in\Gamma} z_\gamma \log\Big(\frac{\theta^{k+1}_\gamma}{\theta^k_\gamma}\Big) - \sum_{\gamma\in\Gamma} \theta^{k+1}_\gamma \log\Big(\frac{\theta^{k+1}_\gamma}{\theta^k_\gamma}\Big) \\
&= \sum_{\gamma\in\Gamma} z_\gamma \log\Big(\frac{z_\gamma}{\theta^k_\gamma}\Big) - \sum_{\gamma\in\Gamma} z_\gamma \log\Big(\frac{z_\gamma}{\theta^{k+1}_\gamma}\Big) + \\
&\quad - \sum_{\gamma\in\Gamma} \theta^{k+1}_\gamma \log\Big(\frac{\theta^{k+1}_\gamma}{\theta^k_\gamma}\Big) \\
&= D_{KL}(z||\theta^k) - D_{KL}(z||\theta^{k+1}) - D_{KL}(\theta^{k+1}||\theta^k)
\end{aligned}$$

Therefore, we can rewrite (A.5) as

$$\begin{aligned}
\eta^{-1}(\phi(\theta^{k+1}) - \phi(z)) \;\leq\; & D_{KL}(z||\theta^k) - D_{KL}(z||\theta^{k+1}) + \\
& - D_{KL}(\theta^{k+1}||\theta^k) .
\end{aligned}$$

We now consider an arbitrary z in $\mathcal{X}^*$. Hence, since $\phi(\theta^{k+1}) - \phi(z) \geq 0$ and D_{KL} is always non-negative, the equation above yields

$$\begin{aligned}
0 \;&\leq\; \eta^{-1}(\phi(\theta^{k+1}) - \phi(z)) + D_{KL}(\theta^{k+1}||\theta^k) \\
&\leq\; D_{KL}(z||\theta^k) - D_{KL}(z||\theta^{k+1}) ,
\end{aligned} \tag{A.6}$$

Hence, the sequence $\{D_{KL}(z\|\theta^k)\}_{k\geq 0}$ is non-increasing and non-negative, and therefore convergent. This, together with (A.6), implies that

$$\lim_{k\to+\infty} D_{KL}(\theta^{k+1}||\theta^k) = 0 ,$$

$$l = \lim_{k\to+\infty} \phi(\theta^k) = \phi(z) = \min_{x\in\Delta} \phi(x) .$$

This proves that θ^k approaches the set $\mathcal{X}^*$ as k grows large. We now aim to prove that θ^k admits a limit. Since Δ is compact, there exists a subsequence $\{\theta^{k_j}\}_{j\geq 0}$ and $\bar{z} \in \Delta$ such that θ^{k_j} converges to $\bar{z}$. By continuity of ϕ, $\phi(\bar{z}) = \ell = \min_{x\in\Delta} \phi(x)$, i.e., $\bar{z} \in \mathcal{X}^*$. Then, (A.6) holds true with $z = \bar{z}$, which implies by the same arguments used before that $D_{KL}(\bar{z}\|\theta^k)$ is a convergent sequence. However, since θ^{k_j} converges to $\bar{z}$ as j grows large, we have

$$D_{KL}(\bar{z}\|\theta^{k_j}) \xrightarrow{j\to+\infty} 0 ,$$

and therefore also

$$D_{KL}(\bar{z}\|\theta^k) \xrightarrow{k\to+\infty} 0 .$$

This in turn implies by standard properties of D_{KL} (Pinsker's inequality) that

$$\|\theta^k - \bar{z}\|_1 \leq \sqrt{2\, D_{KL}(\bar{z}\|\theta^k)} \xrightarrow[k\to+\infty]{} 0, \tag{A.7}$$

which proves that $\theta^k \to \bar{z} \in \mathcal{X}^*$. ∎

Appendix B. PRELIMINARY LEMMAS

In this section, we introduce useful lemmas to prove the main result of this paper. In particular, Lemma 1 provides an explicit characterization of the invariant distribution π_n^θ of the noisy best response, while Lemma 2 proves that, as the population grows large, the invariant distribution concentrates around $\Psi(\theta)$. Lemma 3 then shows that, as the number of players gets large, the expected total travel time under the invariant distribution π_n^θ gets arbitrarily close to the total travel time evaluated at $\Psi(\theta)$. Finally, Lemmas 4-6 state that, as the population grows large, the discrete dynamical system (5) approximates the limit system (A.1).

Lemma 1. The Markov Chain $X(t)$ with tolls (4) admits the following unique invariant distribution

$$\pi_n^\theta(x) = \frac{\binom{n}{nx}\prod_{a\in\Gamma}\theta_a^{nx_a}e^{-\eta^{-1}\bar\phi_n(x)}}{\sum_{y\in\Delta_n}\binom{n}{ny}\prod_{a\in\Gamma}\theta_a^{ny_a}e^{-\eta^{-1}\bar\phi_n(y)}}\,, \tag{B.1}$$

where, for every x in Δ_n,

$$\bar\phi_n(x) = \sum_{e\in\mathcal{E}}\sum_{s=0}^{n(Ax)_e}\Big(\tau_e\Big(\frac{s}{n}\Big)+\tilde\omega_e\Big(\frac{s}{n}\Big)\Big)\,.$$

Proof. It follows from Monderer and Shapley (1996) that the game admits a potential function, i.e., there exists a function $\phi_n:\Delta_n\to\mathbb{R}$ such that, given x,y in Δ_n with $x = y+\frac{1}{n}(u_i-u_j)$,

$$\phi_n(x)-\phi_n(y) = c_i^\theta(x)-c_j^\theta(y)\,. \tag{B.2}$$

In particular, the potential is given by

$$\begin{aligned}
\phi_n(x) &= \sum_{e\in\mathcal{E}}\sum_{s=0}^{n(Ax)_e}c_e^\theta\Big(\frac{s}{n}\Big) = \sum_{e\in\mathcal{E}}\sum_{s=0}^{n(Ax)_e}\Big(\tau_e\Big(\frac{s}{n}\Big)+\omega_e^\theta\Big(\frac{s}{n}\Big)\Big)\\
&= \sum_{e\in\mathcal{E}}\sum_{s=0}^{n(Ax)_e}\Big(\tau_e\Big(\frac{s}{n}\Big)+\tilde\omega_e\Big(\frac{s}{n}\Big)+\bar\omega_e(\theta)\Big)\\
&= \bar\phi_n(x)+n\sum_{e\in\mathcal{E}}(Ax)_e\,\bar\omega_e(\theta)\\
&= \bar\phi_n(x)+n\sum_{\gamma\in\Gamma}x_\gamma\sum_{e\in\gamma}\bar\omega_e(\theta)\\
&= \bar\phi_n(x)-n\eta\sum_{\gamma\in\Gamma}x_\gamma\log\theta_\gamma\,.
\end{aligned}$$

Then, consider two states x and y in Δ_n such that $x = y+\frac{1}{n}(u_i-u_j)$ and note that

$$\begin{aligned}
\frac{P_{yx}^{(n)}}{P_{xy}^{(n)}} &= \frac{y_j e^{-\eta^{-1}c_i^\theta(x)}}{x_i e^{-\eta^{-1}c_j^\theta(y)}}\frac{\sum_{a\in\Gamma}e^{-\eta^{-1}\left(c_a^\theta\left(x+\frac{1}{n}(u_a-u_i)\right)\right)}}{\sum_{a\in\Gamma}e^{-\eta^{-1}\left(c_a^\theta\left(y+\frac{1}{n}(u_a-u_j)\right)\right)}}\\
&= \frac{y_j}{x_i}\frac{e^{-\eta^{-1}\phi_n(x)}}{e^{-\eta^{-1}\phi_n(y)}} = \frac{y_j}{x_i}\frac{\prod_{a\in\Gamma}\theta_a^{nx_a}e^{-\eta^{-1}\bar\phi_n(x)}}{\prod_{a\in\Gamma}\theta_a^{ny_a}e^{-\eta^{-1}\bar\phi_n(y)}}\\
&= \frac{y_j}{x_i}\frac{\theta_i e^{-\eta^{-1}\bar\phi_n(x)}}{\theta_j e^{-\eta^{-1}\bar\phi_n(y)}}\,,
\end{aligned} \tag{B.3}$$

where the second equality follows from (B.2) and the fact that

$$\begin{aligned}
x+\frac{1}{n}(u_a-u_i) &= y+\frac{1}{n}(u_i-u_j)+\frac{1}{n}(u_a-u_i)\\
&= y+\frac{1}{n}(u_a-u_j)\,,
\end{aligned}$$

so that the sums in the numerator and the denominator simplify, the third one from the properties of logarithms, and the fourth one from $x = y+\frac{1}{n}(u_i-u_j)$. We now prove that

$$\frac{\pi_n^\theta(x)}{\pi_n^\theta(y)} = \frac{P_{yx}^{(n)}}{P_{xy}^{(n)}}\,,$$

that is, the Markov chain is reversible with invariant distribution π_n^θ. A direct check shows that this condition is verified with

$$\pi_n^\theta(x) = \frac{\binom{n}{nx}\prod_{a\in\Gamma}\theta_a^{nx_a}e^{-\eta^{-1}\bar\phi_n(x)}}{\sum_{z\in\Delta_n}\binom{n}{nz}\prod_{a\in\Gamma}\theta_a^{nz_a}e^{-\eta^{-1}\bar\phi_n(z)}}\,.$$

This follows from the fact that for every x,y in Δ_n such that $x = y+\frac{1}{n}(u_i-u_j)$ it holds that

$$\begin{aligned}
\frac{\pi_n^\theta(x)}{\pi_n^\theta(y)} &= \frac{\binom{n}{nx}\prod_{a\in\Gamma}\theta_a^{nx_a}e^{-\eta^{-1}\bar\phi_n(x)}}{\binom{n}{ny}\prod_{a\in\Gamma}\theta_a^{ny_a}e^{-\eta^{-1}\bar\phi_n(y)}}\\
&= \frac{\frac{n!}{(nx_1)!,\dots,(nx_{|\Gamma|})!}\prod_{a\in\Gamma}\theta_a^{nx_a}e^{-\eta^{-1}\bar\phi_n(x)}}{\frac{n!}{(ny_1)!,\dots,(ny_{|\Gamma|})!}\prod_{a\in\Gamma}\theta_a^{ny_a}e^{-\eta^{-1}\bar\phi_n(y)}}\\
&= \frac{(ny_i)!(ny_j)!}{(nx_i)!(nx_j)!}\frac{\theta_i^{nx_i}\theta_j^{nx_j}}{\theta_i^{ny_i}\theta_j^{ny_j}}\frac{e^{-\eta^{-1}\bar\phi_n(x)}}{e^{-\eta^{-1}\bar\phi_n(y)}}\\
&= \frac{y_j}{x_i}\frac{\theta_i}{\theta_j}\frac{e^{-\eta^{-1}\bar\phi_n(x)}}{e^{-\eta^{-1}\bar\phi_n(y)}}\,,
\end{aligned}$$

where the third and fourth equalities follows from the fact that $x = y+\frac{1}{n}(u_i-u_j)$. Notice that this is exactly equal to (B.3), thus concluding the proof. ■

The next result shows that, as n grows large, the invariant distribution π_n^θ concentrates around $\Psi(\theta)$. To this, let us introduce the ρ-neighborhood of $\Psi(\theta)$ by

$$U_{\rho,n}(\theta) = \Big\{x\in\Delta_n : ||x-\Psi(\theta)||_\infty<\rho\Big\}\,. \tag{B.4}$$

and define its complement set, i.e.,

$$U_{\rho,n}^c(\theta) = \Delta_n\setminus U_{\rho,n}(\theta)\,.$$

Moreover, for every set M we define the invariant distribution of M as

$$\pi_n^\theta(M) := \sum_{x\in M}\pi_n^\theta(x)\,.$$

Lemma 2. For every compact $K\subset\mathrm{int}(\Delta)$, for every $\rho>0$, and for every $\epsilon>0$, there exists $\hat n_{\epsilon,K,\rho}$ such that

$$\sup_{\theta\in K}\pi_n^\theta(U_{\rho,n}^c(\theta))<\epsilon$$

for every $n>\hat n_{\epsilon,K,\rho}$.

Proof. Recall from Lemma 1 that the invariant distribution is in the form (B.1), and notice that it can be equivalently written as

$$\pi_n^\theta(x) = \frac{\exp(B_n^\theta(x))}{Z_n^\theta}\,,$$

where

$$B_n^\theta(x) = \sum_{k=1}^{n}\log k - \sum_{i=1}^{|\Gamma|}\sum_{k=1}^{nx_i}\log k + n\sum_{a\in\Gamma}x_a\log(\theta_a) + \\ - \eta^{-1}\bar{\phi}_n(x)$$

and

$$Z_n^\theta = \sum_{x\in\Delta_n}\exp\big(B_n^\theta(x)\big)\,.$$

The proof proceeds in two steps. First, we show that $\frac{1}{n}B_n^\theta(x)$ converges uniformly to $-F^\theta(x)$, where $F^\theta(x) = \eta^{-1}\phi(x) + D_{KL}(x\|\theta)$ and then we prove that the invariant distribution π_n^θ concentrates in a neighborhood of $\Psi(\theta)$. So, first consider

$$\begin{aligned}
&\Big|\frac{1}{n}B_n^\theta(x) + F^\theta(x)\Big| \\
&\le \Big|\frac{1}{n}\Big(\sum_{k=1}^{n}\log k - \sum_{a\in\Gamma}\sum_{k=1}^{nx_a}\log k\Big) + \sum_{a\in\Gamma}x_a\log x_a\Big| \\
&+ \Big|\sum_{a\in\Gamma}x_a\log\theta_a - \sum_{a\in\Gamma}x_a\log\theta_a\Big| \\
&+ \Big|\frac{\eta^{-1}}{n}\bar{\phi}_n(x) - \eta^{-1}\phi(x)\Big| \\
&= \Big|\frac{1}{n}\Big(\sum_{k=1}^{n}\log k - \sum_{a\in\Gamma}\sum_{k=1}^{nx_a}\log k\Big) + \sum_{a\in\Gamma}x_a\log x_a\Big| \\
&+ \Big|\frac{\eta^{-1}}{n}\bar{\phi}_n(x) - \eta^{-1}\phi(x)\Big|\,.
\end{aligned}$$

We start by bounding the first term, by finding an upper bound on the binomial $\binom{n}{nx}$. Since $\sum_{a\in\Gamma}x_a = 1$, we know that

$$\sum_{y\in\Delta_n}\binom{n}{ny}\prod_{a\in\Gamma}x_a^{ny_a} = 1\,. \tag{B.5}$$

All the terms in (B.5) are nonnegative, so it follows that

$$\begin{aligned}
\binom{n}{nx}\prod_{a\in\Gamma}x_a^{nx_a} &= \binom{n}{nx}\prod_{a\in\Gamma}e^{nx_a\log(x_a)} \\
&= \binom{n}{nx}e^{n\sum_{a\in\Gamma}x_a\log x_a} \le 1
\end{aligned}$$

which yields

$$\binom{n}{nx} \le e^{-n\sum_{a\in\Gamma}x_a\log x_a}\,. \tag{B.6}$$

We proceed to then found a lower bound on $\binom{n}{nx}$. To this end, for every y in Δ_n consider

$$\begin{aligned}
\frac{\binom{n}{ny}\prod_{a\in\Gamma}x_a^{ny_a}}{\binom{n}{nx}\prod_{a\in\Gamma}x_a^{nx_a}} &= \frac{\binom{n}{ny}\prod_{a\in\Gamma:x_a>0}x_a^{ny_a}}{\binom{n}{nx}\prod_{a\in\Gamma:x_a>0}x_a^{nx_a}} \\
&= \prod_{a\in\Gamma:x_a>0}\frac{(nx_a)!}{(ny_a)!}x_a^{ny_a-nx_a} \\
&\le \prod_{a\in\Gamma:x_a>0}(nx_a)^{nx_a-ny_a}x_a^{ny_a-nx_a} \\
&= \prod_{a\in\Gamma:x_a>0}n^{nx_a-ny_a} = n^0 = 1\,,
\end{aligned}$$

where the inequality follows from the standard bound $\frac{m!}{l!} \le l^{m-l}$, see (Cover and Thomas, 1991, Proof of Theorem 11.1.3). From the inequality above, it follows that

$$\binom{n}{ny}\prod_{a\in\Gamma}x_a^{ny_a} \le \binom{n}{nx}\prod_{a\in\Gamma}x_a^{nx_a}\,.$$

Summing this bound over all y in Δ_n, we obtain

$$\begin{aligned}
1 &= \sum_{y\in\Delta_n}\binom{n}{ny}\prod_{a\in\Gamma}x_a^{ny_a} \le \sum_{y\in\Delta_n}\binom{n}{nx}\prod_{a\in\Gamma}x_a^{nx_a} \\
&= |\Delta_n|\binom{n}{nx}\prod_{a\in\Gamma}x_a^{nx_a} = |\Delta_n|\binom{n}{nx}e^{n\sum_{a\in\Gamma}x_a\log x_a} \\
&\le (n+1)^{|\Gamma|}\binom{n}{nx}e^{n\sum_{a\in\Gamma}x_a\log x_a}\,,
\end{aligned}$$

Here, the first inequality holds because the summand no longer depends on y; the second follows from the fact that $|\Delta_n| \le (n+1)^{|\Gamma|}$, see (Cover and Thomas, 1991, Theorem 11.1.1). Rearranging terms, we conclude that

$$\binom{n}{nx} \ge \frac{e^{-n\sum_{a\in\Gamma}x_a\log x_a}}{(n+1)^{|\Gamma|}}\,. \tag{B.7}$$

Combining (B.7) and (B.6) together, we obtain

$$\frac{e^{-n\sum_{a\in\Gamma}x_a\log x_a}}{(n+1)^{|\Gamma|}} \le \binom{n}{nx} \le e^{-n\sum_{a\in\Gamma}x_a\log x_a}\,,$$

to which we can apply logarithms to achieve

$$\begin{aligned}
-n\sum_{a\in\Gamma}x_a\log x_a - |\Gamma|\log(n+1) &\le \log\Big(\binom{n}{nx}\Big) \\
&\le -n\sum_{a\in\Gamma}x_a\log x_a\,.
\end{aligned}$$

Dividing by n and moving $-n\sum_{a\in\Gamma}x_a\log x_a$ to the the middle term, we obtain

$$-\frac{|\Gamma|\log(n+1)}{n} \le \frac{1}{n}\log\Big(\binom{n}{nx}\Big) + \sum_{a\in\Gamma}x_a\log x_a \le 0\,,$$

or equivalently

$$\begin{aligned}
&\Big|\frac{1}{n}\Big(\sum_{k=1}^{n}\log k - \sum_{a\in\Gamma}\sum_{k=1}^{nx_a}\log k\Big) + \sum_{a\in\Gamma}x_a\log x_a\Big| \\
&\quad\le \frac{|\Gamma|\log(n+1)}{n}\,.
\end{aligned}$$

Since the bound does not depend on x in Δ_n we conclude that

$$\begin{aligned}
&\sup_{x\in\Delta_n}\Big|\frac{1}{n}\Big(\sum_{k=1}^{n}\log k - \sum_{a\in\Gamma}\sum_{k=1}^{nx_a}\log k\Big) + \sum_{a\in\Gamma}x_a\log x_a\Big| \\
&\quad\le \frac{|\Gamma|\log(n+1)}{n}\,,
\end{aligned} \tag{B.8}$$

which tends to 0 uniformly as n grows large.

We are then left to prove that $\frac{\eta^{-1}}{n}\bar{\phi}_n(x)$ converges uniformly to $\eta^{-1}\phi(x)$. First, consider a sequence $(k_n)_n$ such that k_n/n belongs to Δ_n for every n, and $k_n/n \to x$, and note that

$$\begin{aligned}\lim_{n\to+\infty}\frac{1}{n}\bar{\phi}_n\Big(\frac{k_n}{n}\Big) &= \lim_{n\to+\infty}\frac{1}{n}\sum_{e\in\mathcal{E}}\Big(\sum_{j=0}^{(Ak_n)_e}\tau_e\Big(\frac{j}{n}\Big)+\tilde{\omega}_e\Big(\frac{j}{n}\Big)\Big)\\ &= \sum_{e\in\mathcal{E}}\int_0^{(Ax)_e}(\tau_e(s)+\tilde{\omega}_e(s))\,ds\\ &= \sum_{e\in\mathcal{E}}\int_0^{(Ax)_e}\left(\frac{d}{ds}\left(s\tau_e(s)\right)\right)ds\\ &= \sum_{e\in\mathcal{E}}(Ax)_e\tau_e\left((Ax)_e\right)\\ &= \sum_{e\in\mathcal{E}}\sum_{\gamma\in\Gamma}A_{e\gamma}x_\gamma\tau_e\left((Ax)_e\right)\\ &= \sum_{\gamma\in\Gamma}x_\gamma\sum_{e\in\mathcal{E}}A_{e\gamma}\tau_e\left((Ax)_e\right)\\ &= \phi(x)\,,\end{aligned} \tag{B.9}$$

where the third equality follows from the fact that

$$\frac{d}{ds}(s\tau_e(s)) = \tau_e(s)+s\tau_e'(s) = \tau_e(s)+\tilde{\omega}_e(s)\,.$$

Since τ_e and $\tilde{\omega}_e$ are continuous on $[0,1]$, they are also uniformly continuous and bounded. Let $M_e := \max_{s\in[0,1]}\left|\tau_e(s)+\tilde{\omega}_e(s)\right|$ and

$$g_e(\nu) := \sup_{\substack{s,t\in[0,1]\\|s-t|\le\nu}}\left|\tau_e(s)+\tilde{\omega}_e(s)-\tau_e(t)-\tilde{\omega}_e(t)\right|,$$

which satisfies $g_e(\nu)\to 0$ as $\nu\to 0$ by uniform continuity of $\tau_e+\tilde{\omega}_e$. For $x=k/n\in\Delta_n$, since $(Ak)_e/n=(Ax)_e\le 1$, we have

$$\begin{aligned}&\Big|\frac{1}{n}\sum_{j=0}^{(Ak)_e}\Big(\tau_e\Big(\frac{j}{n}\Big)+\tilde{\omega}_e\Big(\frac{j}{n}\Big)\Big)-\int_0^{(Ax)_e}\left(\tau_e(s)+\tilde{\omega}_e(s)\right)ds\Big|\\ &= \Big|\sum_{j=0}^{(Ak)_e-1}\int_{\frac{j}{n}}^{\frac{j+1}{n}}\Big(\tau_e\Big(\frac{j}{n}\Big)+\tilde{\omega}_e\Big(\frac{j}{n}\Big)-\tau_e(s)-\tilde{\omega}_e(s)\Big)\,ds\\ &\quad+\frac{1}{n}\Big(\tau_e\big((Ax)_e\big)+\tilde{\omega}_e\big((Ax)_e\big)\Big)\Big|\\ &\le \sum_{j=0}^{(Ak)_e-1}\int_{\frac{j}{n}}^{\frac{j+1}{n}}\Big|\tau_e\Big(\frac{j}{n}\Big)+\tilde{\omega}_e\Big(\frac{j}{n}\Big)-\tau_e(s)-\tilde{\omega}_e(s)\Big|\,ds+\frac{M_e}{n}\\ &\le \sum_{j=0}^{(Ak)_e-1}\int_{\frac{j}{n}}^{\frac{j+1}{n}}g_e\Big(\frac{1}{n}\Big)\,ds+\frac{M_e}{n} \;=\; \frac{(Ak)_e}{n}\,g_e\Big(\frac{1}{n}\Big)+\frac{M_e}{n}\\ &\le g_e\Big(\frac{1}{n}\Big)+\frac{M_e}{n}\,.\end{aligned}$$

The first equality follows by isolating the term $j=(Ak)_e$ of the sum, splitting $[0,(Ax)_e]$ into the $(Ak)_e$ intervals $[j/n,(j+1)/n]$, and writing $\frac{1}{n}\big(\tau_e(\frac{j}{n})+\tilde{\omega}_e(\frac{j}{n})\big) = \int_{j/n}^{(j+1)/n}\big(\tau_e(\frac{j}{n})+\tilde{\omega}_e(\frac{j}{n})\big)\,ds$. The first inequality is the triangle inequality, together with $\left|\tau_e((Ax)_e)+\tilde{\omega}_e((Ax)_e)\right|\le M_e$, which holds since $(Ax)_e\in[0,1]$. The second inequality follows from the definition of g_e, since $|s-j/n|\le 1/n$ for every $s\in[j/n,(j+1)/n]$. For the last inequality, we used the fact that $(Ak)_e/n=(Ax)_e\le 1$.

Summing over $e\in\mathcal{E}$, and recalling that $\frac{1}{n}\bar{\phi}_n(x)$ and $\phi(x)$ are the sums over e of the two terms on the left-hand side (cf. (B.9)), the triangle inequality gives

$$\sup_{x\in\Delta_n}\left|\frac{1}{n}\bar{\phi}_n(x)-\phi(x)\right| \le \sum_{e\in\mathcal{E}}\Big(g_e\Big(\frac{1}{n}\Big)+\frac{M_e}{n}\Big)\xrightarrow{n\to+\infty}0\,, \tag{B.10}$$

since the bound does not depend on x and $g_e(1/n)\to 0$ for every e.

Combining (B.8) and (B.10) with the decomposition of $\left|\frac{1}{n}B_n^\theta(x)+F^\theta(x)\right|$ above, we obtain

$$\begin{aligned}\sup_{x\in\Delta_n}\left|\frac{1}{n}B_n^\theta(x)+F^\theta(x)\right| \le\; &\frac{|\Gamma|\log(n+1)}{n}\\ &+\eta^{-1}\sum_{e\in\mathcal{E}}\Big(g_e\Big(\frac{1}{n}\Big)+\frac{M_e}{n}\Big)\,.\end{aligned} \tag{B.11}$$

Since each term on the right-hand side vanishes as n grows large, we conclude that $\frac{1}{n}B_n^\theta$ converges uniformly to $-F^\theta$ on Δ_n.

We are then left to prove that the invariant distribution concentrates around a neighborhood of $\Psi(\theta)$. First, by construction, since $F^\theta(x)$ is a strictly convex function,

$$\Psi(\theta) = \arg\min_{x\in\Delta}\{F^\theta(x)\}\,,$$

is its unique minimizer. Hence, for every $\rho>0$ and compact $K\subset\mathrm{int}(\Delta)$, there exists a constant $c_{\rho,K}>0$ such that

$$F^\theta(x)\ge F^\theta(\Psi(\theta))+c_{\rho,K}\,,\ \forall x\in U_{\rho,n}^c(\theta)\,,\ \forall\theta\in K\,, \tag{B.12}$$

i.e., $F^\theta(x)$ computed at every point x in $U_{\rho,n}^c(\theta)$ is strictly larger than $F^\theta(\Psi(\theta))$. Moreover, from (B.11) it follows that there exists $n_{\rho,K}$ such that for every $n>n_{\rho,K}$, x in Δ_n, and θ in K it holds that

$$-\frac{c_{\rho,K}}{4}\le\frac{1}{n}B_n^\theta(x)+F^\theta(x)\le\frac{c_{\rho,K}}{4}\,. \tag{B.13}$$

By considering the upper bound of (B.13) and applying (B.12) we obtain

$$\begin{aligned}B_n^\theta(x) &\le -n\Big(F^\theta(\Psi(\theta))+c_{\rho,K}-\frac{c_{\rho,K}}{4}\Big)\\ &= -n\Big(F^\theta(\Psi(\theta))+\frac{3c_{\rho,K}}{4}\Big)\end{aligned}$$

which implies that

$$\begin{aligned}\sum_{x\in U_{\rho,n}^c(\theta)}e^{B_n^\theta(x)} &\le |\Delta_n|e^{-n\left(F^\theta(\Psi(\theta))+\frac{3c_{\rho,K}}{4}\right)}\\ &\le (n+1)^{|\Gamma|}e^{-n\left(F^\theta(\Psi(\theta))+\frac{3c_{\rho,K}}{4}\right)}\,.\end{aligned} \tag{B.14}$$

We now aim to lower bound Z_n^θ. To this end, consider a sequence $(x_n(\theta))_{n\in\mathbb{N}}$ in Δ_n such that

$$\sup_{\theta\in K}||x_n(\theta)-\Psi(\theta)||\to 0$$

and consider the lower bound of Equation (B.13), i.e.,

$$B_n^\theta(x_n(\theta))\ge -n\Big(F^\theta(x_n(\theta))+\frac{c_{\rho,K}}{4}\Big)\,.$$

By uniform continuity of F^θ, there exists $\bar{n}_{\rho,K}$ such that

$$\sup_{\theta\in K}\left|F^\theta(x_n(\theta))-F^\theta(\Psi(\theta))\right|\le c_{\rho,K}/4$$

for every $n>\bar{n}_{\rho,K}$. This implies that

$$\begin{aligned}B_n^\theta(x_n(\theta)) &\ge -n(F^\theta(\Psi(\theta))+\frac{c_{\rho,K}}{4}+\frac{c_{\rho,K}}{4})\\ &= -n(F^\theta(\Psi(\theta))+\frac{c_{\rho,K}}{2})\,.\end{aligned}$$

From this it follows that

$$Z_n^\theta \geq e^{B_n^\theta(x_n(\theta))} \geq e^{-n\left(F^\theta(\Psi(\theta))+\frac{c_{\rho,K}}{2}\right)} . \qquad \text{(B.15)}$$

By combining (B.14) and (B.15), for every $n > \max\{n_{\rho,K}, \bar{n}_{\rho,K}\}$,

$$\sup_{\theta\in K} \pi_n^\theta\big(U^c_{\rho,n}(\theta)\big) = \sup_{\theta\in K} \frac{\sum_{x\in U^c_{\rho,n}(\theta)} e^{B_n^\theta(x)}}{Z_n^\theta} \leq (n+1)^{|\Gamma|} e^{-n\frac{c_{\rho,K}}{4}}.$$

Since the right-hand side does not depend on θ and vanishes as $n \to +\infty$, there exists $n'_{\epsilon,\rho,K}$ such that it is smaller than ϵ for every $n > n'_{\epsilon,\rho,K}$. The claim follows with $\hat{n}_{\epsilon,K,\rho} := \max\{n_{\rho,K}, \bar{n}_{\rho,K}, n'_{\epsilon,\rho,K}\}$. ■

As a consequence of this result, the expected total travel time computed with respect to the invariant distribution π_n^θ converges uniformly in θ to the expected total travel time computed with respect to a delta function centered in $\Psi(\theta)$, as the next result shows.

Lemma 3. For every compact $K \subset \text{int}(\Delta)$ and for every $\epsilon > 0$, there exists $\tilde{n}_{\epsilon,K}$ such that

$$\sup_{\theta\in K} |\mathbb{E}_{\pi_n^\theta}[\phi(X)] - \mathbb{E}_{\delta^{\Psi(\theta)}}[\phi(X)]| < \epsilon$$

for every $n > \tilde{n}_{\epsilon,K}$.

Proof. First, from the triangular inequality it holds that, for every $\rho > 0$,

$$\begin{aligned}
&\sup_{\theta\in K} |\mathbb{E}_{\pi_n^\theta}[\phi(X)] - \mathbb{E}_{\delta^{\Psi(\theta)}}[\phi(X)]| \\
&= \sup_{\theta\in K} \Big| \sum_{x\in\Delta_n} \pi_n^\theta(x)\big(\phi(x) - \phi(\Psi(\theta))\big)\Big| \\
&\leq \sup_{\theta\in K} \Big(\sup_{x\in U_{\rho,n}(\theta)} |\phi(x) - \phi(\Psi(\theta))| + 2||\phi||_\infty \pi_n^\theta(U^c_{\rho,n}(\theta))\Big) .
\end{aligned}$$

We now aim to upper bound the two terms in the external superior. Since ϕ is continuous, for every K and for every $\epsilon > 0$, there exists ρ_ϵ such that

$$\sup_{\theta\in K} \sup_{x\in U_{\rho_\epsilon,n}(\theta)} \Big\{|\phi(x) - \phi(\Psi(\theta))\Big\} < \frac{\epsilon}{2} .$$

By Lemma 2 applied with $\epsilon/(4\|\phi\|_\infty)$ and $\rho = \rho_\epsilon$,

$$2\|\phi\|_\infty \sup_{\theta\in K} \pi_n^\theta\big(U^c_{\rho_\epsilon,n}(\theta)\big) < \frac{\epsilon}{2}$$

for every $n > \tilde{n}_{\epsilon,K} := \hat{n}_{\epsilon/(4\|\phi\|_\infty),K,\rho_\epsilon}$. Summing the two contributions we obtain the statement. ■

We now prove that $\psi_n(\theta)$ converges uniformly to $\Psi(\theta)$.

Lemma 4. For every compact $K \subset \text{int}(\Delta)$ and for every $\epsilon > 0$, there exists $\bar{n}_{\epsilon,K}$ such that

$$\sup_{\theta\in K} \|\psi_n(\theta) - \Psi(\theta)\| < \epsilon \qquad \text{(B.16)}$$

for every $n > \bar{n}_{\epsilon,K}$.

Proof. From Lemma 2, the invariant distributions π_n^θ concentrates around $\Psi(\theta)$ as n grows large. Namely, for every compact $K \subset \text{int}(\Delta)$, for every $\rho > 0$, and for every $\epsilon > 0$, there exists $\hat{n}_{\epsilon,K,\rho}$ such that

$$\sup_{\theta\in K} \pi_n^\theta(U^c_{\rho,n}(\theta)) < \epsilon.$$

Given a fixed ϵ, choose $\rho < \epsilon/2$ and consider

$$\begin{aligned}
\sup_{\theta\in K} \|\psi_n(\theta) - \Psi(\theta)\|_\infty &= \sup_{\theta\in K} \Big\| \sum_{x\in\Delta_n} (x - \Psi(\theta))\pi_n^\theta(x)\Big\|_\infty \\
&\leq \sup_{\theta\in K} \sum_{x\in U_{\rho,n}(\theta)} \|x - \Psi(\theta)\|_\infty \pi_n^\theta(x) \\
&\quad + \sup_{\theta\in K} \sum_{x\in U^c_{\rho,n}(\theta)} \|x - \Psi(\theta)\|_\infty \pi_n^\theta(x) \\
&< \rho + \sup_{\theta\in K} \pi_n^\theta(U^c_{\rho,n}(\theta)) \\
&< \frac{\epsilon}{2} + \sup_{\theta\in K} \pi_n^\theta(U^c_{\rho,n}(\theta)) ,
\end{aligned}$$

where the first step follows from the definition of $\psi_n(\theta)$ and the fact that $\sum_{x\in\Delta_n} \pi_n^\theta(x) = 1$. The second step follows by splitting the sum into a sum of the elements inside and outside the set $U_{\rho,n}(\theta)$, respectively. Lastly, the third steps follows from the definition of $U_{\rho,n}(\theta)$ (B.4) and the worst-case bound. Since $\rho < \epsilon/2$ and by applying Lemma 2 with $\epsilon/2$ the claim follows. ■

We now introduce some notation. In particular, we first define the closure of the ball of radius χ around a point $\bar{\theta}$ as

$$B_\chi(\bar{\theta}) = \{\theta \in \Delta : ||\theta - \bar{\theta}|| \leq \chi\} , \qquad \text{(B.17)}$$

and then we define the compact set

$$K_{k,\theta^0,\chi} := \bigcup_{r=0}^{k} B_\chi(\bar{\theta}^r) , \qquad \text{(B.18)}$$

which is the union of the balls around the trajectory of the limit map (A.1) within k steps.

The next lemma shall prove that, for a fixed k, there exists a sufficiently large n such that the trajectory of the finite-population dynamical system (5) also belongs to $K_{k,\theta^0,\chi}$.

Lemma 5. For every $k \geq 0$ and for every θ^0 in $\text{int}(\Delta)$, there exists $\check{n}_{k,\theta^0}$ such that

$$\psi_n^h(\theta^0) \in K_{k,\theta^0,\chi} , \ \forall h = 0, \dots, k ,$$

for every $n > \check{n}_{k,\theta^0,\chi}$.

Proof. Let

$$\bar{\theta}^r := \Psi^r(\bar{\theta}^0), \qquad r = 0, \dots, k.$$

Since the finite set $\{\bar{\theta}^0, \dots, \bar{\theta}^k\}$ is contained in $\text{int}(\Delta)$, there exists χ such that we define the compact set as in (B.18).

We prove by induction that, for every $r = 0, \dots, k$ and every $\epsilon > 0$, there exists $n_{r,\epsilon}$ such that

$$\|\psi_n^r(\theta^0) - \Psi^r(\theta^0)\| < \epsilon , \qquad \forall\, n > n_{r,\epsilon} . \qquad \text{(B.19)}$$

For $r = 0$ the claim is trivial, since $\psi_n^0(\theta^0) = \Psi^0(\theta^0) = \theta^0$. Suppose that (B.19) holds for some $r < k$, and fix $\epsilon > 0$. Since Ψ is continuous at $\Psi^r(\theta^0)$, there exists $\zeta_\epsilon \in (0, \chi]$ such that, for every $\theta \in \Delta$,

$$\|\theta - \Psi^r(\theta^0)\| < \zeta_\epsilon \implies \|\Psi(\theta) - \Psi(\Psi^r(\theta^0))\| < \frac{\epsilon}{2} . \qquad \text{(B.20)}$$

By the induction hypothesis applied with ζ_ϵ, for every $n > n_{r,\zeta_\epsilon}$ we have

$$\|\psi_n^r(\theta^0) - \Psi^r(\theta^0)\| < \zeta_\epsilon \leq \chi ,$$

so that

$$\psi_n^r(\theta^0) \in B_\chi(\Psi^r(\theta^0)) \subseteq K_{k,\theta^0,\chi} .$$

Moreover, by Lemma 4, since $K_{k,\theta^0,\chi}$ is a compact subset of $\mathrm{int}(\Delta)$, there exists $\bar{n}_{\epsilon,K}$ such that

$$\sup_{\theta\in K_{k,\theta^0}} \|\psi_n(\theta)-\Psi(\theta)\| < \frac{\epsilon}{2}\,, \qquad \forall\, n > \bar{n}_{\epsilon,K}\,. \tag{B.21}$$

Hence, for every $n > n_{r+1,\epsilon} := \max\{n_{r,\zeta_\epsilon}, \bar{n}_{\epsilon,K}\}$, the triangle inequality gives

$$\begin{aligned}\|\psi_n^{r+1}(\theta^0)-\Psi^{r+1}(\theta^0)\| &= \|\psi_n(\psi_n^r(\theta^0))-\Psi(\Psi^r(\theta^0))\| \\ &\le \|\psi_n(\psi_n^r(\theta^0))-\Psi(\psi_n^r(\theta^0))\| + \\ &+ \|\Psi(\psi_n^r(\theta^0))-\Psi(\Psi^r(\theta^0))\| \\ &< \frac{\epsilon}{2}+\frac{\epsilon}{2} = \epsilon\,,\end{aligned}$$

where the first term is bounded using $\psi_n^r(\theta^0)\in K_{k,\theta^0}$ and (B.21), and the second using (B.20). This completes the induction.

Since (B.19) holds for every $\epsilon>0$, we can apply it with $\epsilon=\chi$: for every $h=0,\dots,k$ there exists $n_{h,\chi}$ such that $\|\psi_n^h(\theta^0)-\bar{\theta}^h\|<\chi$ for every $n>n_{h,\chi}$. Setting $\check{n}_{k,\theta^0,\chi} := \max_{h=0,\dots,k} n_{h,\chi}$, we conclude that, for every $n>\check{n}_{k,\theta^0,\chi}$,

$$\psi_n^h(\theta^0)\in B_\chi(\bar{\theta}^h)\subseteq K_{k,\theta^0,\chi}\,, \qquad \forall\, h=0,\dots,k\,.$$

■

The next result proves that, if the k-th iteration of the map Ψ for an initial condition θ^0 is arbitrarily close to x^*, then also the k-th iteration of the map ψ_n is arbitrarily close to x^* for sufficiently large n.

Lemma 6. Consider $k\in\mathbb{N}$ and $\theta^0\in\mathrm{int}(\Delta)$ such that

$$\|\Psi^k(\theta^0)-x^*\|<\varepsilon\,,$$

for some $\varepsilon>0$. Then there exists $\check{n}_{\varepsilon,k,\theta^0}$ in $\mathbb{N}$ such that, for every $n>\check{n}_{\varepsilon,k,\theta^0}$,

$$\|\psi_n^k(\theta^0)-x^*\|<2\varepsilon.$$

Proof. From Lemma 5, we know that for every $n>\check{n}_{k,\theta^0}$ the trajectory ψ_n^h for every $h=0,\dots,k$ is in K_{k,θ^0} and moreover that

$$\psi_n^h(\theta^0)\xrightarrow{n\to+\infty}\Psi^h(\theta^0)\,,\forall h=1,\dots,k\,.$$

Therefore, there also exists $\check{n}_{\epsilon,k,\theta^0}$ in $\mathbb{N}$ such that, for every $n>\check{n}_{\epsilon,k,\theta^0}$,

$$\|\psi_n^k(\theta^0)-\Psi^k(\theta^0)\|<\varepsilon.$$

Using the triangle inequality and combining this inequality with $\|\Psi^k(\theta^0)-x^*\|<\varepsilon$, we obtain

$$\begin{aligned}\|\psi_n^k(\theta^0)-x^*\| &\le \|\psi_n^k(\theta^0)-\Psi^k(\theta^0)\|+\|\Psi^k(\theta^0)-x^*\| \\ &< 2\varepsilon.\end{aligned}$$

This concludes the proof. ■

Lemma 7. Let $m:=\min_{\gamma\in\mathcal{S}} x^*_\gamma$. For every $\theta\in\mathrm{int}(\Delta)$ such that $\|\theta-x^*\|_\infty\le m/2$,

$$\|\Psi(\theta)-x^*\|_1 \;\le\; 2\sqrt{\frac{\|\theta-x^*\|_\infty}{m}}.$$

Proof. Let $\delta := \|\theta-x^*\|_\infty \le m/2$. Since $\theta\in\mathrm{int}(\Delta)$, also $\Psi(\theta)\in\mathrm{int}(\Delta)$ (see Appendix A), so (A.6) holds with $\theta^k=\theta$, $\theta^{k+1}=\Psi(\theta)$ and $z=x^*$. Since $\phi(\Psi(\theta))\ge\phi(x^*)$ and $D_{KL}(\Psi(\theta)\|\theta)\ge 0$, it yields

$$D_{KL}(x^*\|\Psi(\theta))\le D_{KL}(x^*\|\theta).$$

Moreover, for every γ with $x^*_\gamma>0$ we have

$$\theta_\gamma \ge x^*_\gamma-\delta = x^*_\gamma\Big(1-\frac{\delta}{x^*_\gamma}\Big)\ge x^*_\gamma(1-\delta/m)\,,$$

from which it follows that

$$\frac{x^*_\gamma}{\theta_\gamma}\le\frac{1}{1-\frac{\delta}{m}}\,,\implies \log\Big(\frac{x^*_\gamma}{\theta_\gamma}\Big)\le\log\Big(\frac{1}{1-\frac{\delta}{m}}\Big)=-\log\Big(1-\frac{\delta}{m}\Big)\,.$$

Hence, since $-\log(1-t)\le 2t$ for $t\in[0,1/2]$ by convexity,

$$D_{KL}(x^*\|\theta)=\sum_{\gamma:\,x^*_\gamma>0} x^*_\gamma\log\frac{x^*_\gamma}{\theta_\gamma}\le-\log\Big(1-\frac{\delta}{m}\Big)\le\frac{2\delta}{m}.$$

Combining the two bounds with Pinsker's inequality, we obtain

$$\begin{aligned}\|\Psi(\theta)-x^*\|_1 &\le \sqrt{2D_{KL}(x^*\|\Psi(\theta))}\le\sqrt{2D_{KL}(x^*\|\theta)} \\ &\le 2\sqrt{\delta/m}.\end{aligned}$$

Appendix C. PROOF OF THEOREM 1

First, recall that

$$\phi(X)=\sum_{\gamma\in\Gamma}X_\gamma A_{e\gamma}\tau_e\left((AX)_e\right)\,,$$

and that, by Assumption 1, ϕ is strictly convex, so that

$$\mathcal{X}^*=\{x^*\}\,.$$

Let

$$\bar{\theta}^r:=\Psi^r(\bar{\theta}^0),\qquad r=0,\dots,k.$$

Since the finite set $\{\bar{\theta}^0,\dots,\bar{\theta}^k\}$ is contained in $\mathrm{int}(\Delta)$, since the derivative of $D_{KL}(x\|\theta)$ diverges as x approaches the boundary of Δ, we can define the compact set around the trajectory of the limit map, as in (B.18).

By Lemma 5 we know that for every k and for $n>\check{n}_{k,\theta^0}$, the set $\{\psi_n^0(\theta^0),\dots,\psi_n^k(\theta^0)\}$ is in K_{k,θ^0}. From this fact, it follows from Lemma 3 (with $K=K_{k,\theta^0}$ and $\theta=\psi_n^k(\theta^0)$) that, for every $\epsilon>0$, there exists a finite $\tilde{n}_{\epsilon,K_{k,\theta^0}}$ such that

$$\Big|\mathbb{E}_{\pi_n^{\psi_n^k(\theta^0)}}[\phi(X)]-\mathbb{E}_{\delta^{\Psi(\psi_n^k(\theta^0))}}[\phi(X)]\Big|<\frac{\epsilon}{2}\,, \tag{C.1}$$

for every

$$n>\underline{n}_{\epsilon,k,\theta^0}:=\max\{\tilde{n}_{\epsilon,K_{k,\theta^0}},\check{n}_{k,\theta^0}\}\,.$$

From continuity of $\phi(x)$, we have that for every $\epsilon>0$, there exists σ_ϵ such that

$$||x-y||<\sigma_\epsilon\implies|\phi(x)-\phi(y)|<\epsilon/2\,. \tag{C.2}$$

Let $m:=\min_{\gamma\in\Gamma:\,x^*_\gamma>0}x^*_\gamma$ and set

$$\bar{\sigma}_\epsilon:=\min\Big\{\frac{m}{2},\;\frac{m\,\sigma_\epsilon^2}{4}\Big\}.$$

Let $\theta\in\mathrm{int}(\Delta)$ be such that $\delta:=\|\theta-x^*\|_\infty<\bar{\sigma}_\epsilon$. Since $\bar{\sigma}_\epsilon\le m/2$, Lemma 7 applies and, since $\bar{\sigma}_\epsilon\le m\sigma_\epsilon^2/4$ and the square root is increasing,

$$\|\Psi(\theta)-x^*\|_1\le 2\sqrt{\frac{\delta}{m}}<2\sqrt{\frac{\bar{\sigma}_\epsilon}{m}}\le 2\sqrt{\frac{\sigma_\epsilon^2}{4}}=\sigma_\epsilon.$$

Hence, for every $\theta\in\mathrm{int}(\Delta)$,

$$\|\theta-x^*\|_\infty<\bar{\sigma}_\epsilon\implies\|\Psi(\theta)-x^*\|_1<\sigma_\epsilon. \tag{C.3}$$

From Proposition 1, it follows that there exists k_{ϵ,θ^0} such that

$$||\Psi^k(\theta^0)-x^*||<\bar{\sigma}_\epsilon/2\,,\quad\forall k>k_{\epsilon,\theta^0}\,. \tag{C.4}$$

Using (C.4) and Lemma 6, for every $k>k_{\epsilon,\theta^0}$ there exists $\check{n}_{\epsilon,k,\theta^0}$ in $\mathbb{N}$ such that

$$||\psi_n^k(\theta^0)-x^*||<\bar{\sigma}_\epsilon\,,\quad\forall n>\check{n}_{\epsilon,k,\theta^0}\,. \tag{C.5}$$

We can then plug (C.5) into (C.3) (with $\theta = \psi_n^k(\theta^0)$), and get that

$$||\Psi(\psi_n^k(\theta^0)) - \Psi(x^*)|| = ||\Psi(\psi_n^k(\theta^0)) - x^*|| < \sigma_\epsilon\,, \quad \text{(C.6)}$$

where the equality follows from the fact that $x^* = \Psi(x^*)$. From (C.6) and (C.2), we then get that

$$\begin{aligned} &|\mathbb{E}_{\delta^{\Psi(\psi_n^k(\theta^0))}}[\phi(X)] - \mathbb{E}_{\delta^{x^*}}[\phi(X)]| \\ &= |\phi(\Psi(\psi_n^k(\theta^0))) - \phi(x^*)| < \frac{\epsilon}{2}\,, \end{aligned} \quad \text{(C.7)}$$

for every $k > k_{\epsilon,\theta^0}$ and $n > \check{n}_{\epsilon,k,\theta^0}$. Using (C.1), (C.7) and the triangle inequality, we get that

$$\left|\mathbb{E}_{\pi_n^{\psi_n^k(\theta^0)}}[\phi(X)] - \mathbb{E}_{\delta^{x^*}}[\phi(X)]\right| < \epsilon, \quad \text{(C.8)}$$

for every $k > k_\epsilon$ and $n > n^*_{\epsilon,k,\theta^0} := \max(\check{n}_{\epsilon,k,\theta^0}, \underline{n}_{\epsilon,k,\theta^0})$. Recall now that

$$x^* = \arg\min_{x\in\Delta} \phi(x)\,.$$

and that $\theta^k = \psi_n^k(\theta^0)$. Hence, from (C.8)

$$\begin{aligned} \mathbb{E}_{\pi_n^{\theta^k}}[\phi(X)] - \inf_{\pi\in\Delta_n} \mathbb{E}_\pi[\phi(X)] &\leq \mathbb{E}_{\pi_n^{\theta^k}}[\phi(X)] - \mathbb{E}_{\delta^{x^*}}[\phi(X)] \\ &< \epsilon\,, \end{aligned}$$

for every $k > k_{\epsilon,\theta^0}$ and $n > n^*_{\epsilon,k,\theta^0}$. This ends the proof. ■